PREDICTIVE ANALYTICS

Customer Churn Prediction Using Machine Learning

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PREDICTIVE ANALYTICS PROJECT REPORT

Topic: Customer Churn Prediction Using Machine Learning

Subject: Predictive Analytics | Programme: Master of Business Administration | Student: Shruti Singh | United University, Prayagraj | Session: 2025–27

DECLARATION

I hereby declare that the project report entitled **Customer Churn Prediction Using Machine Learning** has been prepared for academic purposes. The project demonstrates the use of predictive analytics techniques to estimate customer churn from selected customer attributes. The dataset is synthetic and has been generated for educational demonstration; it does not represent actual customers or a real organization.

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ABSTRACT

This project demonstrates customer churn prediction using predictive analytics and machine learning. A synthetic dataset of 600 customer records is used to study customer tenure, monthly charges, contract type, internet service, technical support, payment method, usage, complaints, and late payments. Logistic Regression and Random Forest classification models are trained and evaluated on a held-out test set. In this generated sample, 59 customers (9.83%) are labelled as churned. The Random Forest model achieves a ROC-AUC of 0.742, while its recall is 0.133; therefore, the result is useful as a demonstration but not sufficient for real-world deployment. The report explains the predictive workflow, results, limitations, and responsible business recommendations.

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ABBREVIATIONS

Term	Meaning
ML	Machine Learning
KPI	Key Performance Indicator
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
TP	True Positive
FP	False Positive
FN	False Negative
TN	True Negative
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values

1. TITLE AND PROJECT OVERVIEW

1.1 Project Title

Customer Churn Prediction Using Machine Learning

1.2 Area of Study

The project relates to predictive analytics, marketing management, customer relationship management, and machine learning. It focuses on estimating the likelihood that a customer will discontinue a service, based on historical customer characteristics.

1.3 Scope of the Project

• Study selected customer attributes and churn status.

• Prepare a structured dataset for classification modelling.

• Explore churn patterns by tenure and contract type.

• Build Logistic Regression and Random Forest models.

• Evaluate models using accuracy, precision, recall, F1-score, and ROC-AUC.

• Interpret results and outline possible retention actions.

1.4 Objectives

• Understand customer churn and its impact on businesses.

• Explain the role of predictive analytics in customer retention.

• Prepare and analyse a synthetic customer dataset.

• Build classification models to estimate churn likelihood.

• Compare model performance using suitable evaluation metrics.

• Suggest responsible ways to use churn predictions in retention planning.

2. INTRODUCTION

2.1 Background of the Study

Customer retention is important in subscription-based businesses such as telecommunications, internet services, digital platforms, and membership services. When customers discontinue a service, an organization may lose recurring revenue and incur additional costs to acquire replacement customers. Understanding the characteristics associated with churn can help businesses identify customers who may need additional support.

2.2 Meaning of Customer Churn

Customer churn occurs when a customer stops using or paying for a product or service during a defined period. Churn may be voluntary, such as cancellation due to price or dissatisfaction, or involuntary, such as service termination following repeated payment failures. The exact definition should be set consistently before analysis.

2.3 Meaning of Predictive Analytics

Predictive analytics uses historical data, statistical techniques, and machine learning algorithms to estimate the likelihood of future events. In this project, classification models learn patterns from customer attributes and a labelled churn outcome, then estimate churn status for records not used during model training.

2.4 Types of Data Analytics

Type	Purpose	Churn Example
Descriptive	Explains what happened	Summarizing past churn rate
Diagnostic	Explores why it happened	Investigating cancellation reasons
Predictive	Estimates what may happen	Scoring customers for churn risk
Prescriptive	Suggests possible actions	Selecting retention interventions

2.5 Importance of Churn Prediction

• Helps identify customers who may be at risk of leaving.

• Supports proactive customer-service outreach.

• Can help prioritize retention resources.

• Provides a structured way to evaluate customer risk indicators.

• Encourages data-informed customer relationship management.

3. PROBLEM STATEMENT

3.1 Statement of the Problem

Businesses with recurring customer relationships may face customer attrition. Reviewing churn only after customers leave limits the opportunity to respond early. Organizations need methods to examine customer information and estimate which customers may be more likely to discontinue their service.

3.2 Need for the Study

A predictive model can combine several customer attributes and produce a risk estimate. Such estimates may help teams prioritize service reviews, investigate complaints, and design retention programmes. Predictions should support human decision-making rather than automatically determine customer treatment.

3.3 Research Questions

• What proportion of the sample is labelled as churned?

• How does churn vary across contract types and tenure groups?

• Can customer attributes be used to estimate churn status?

• How do Logistic Regression and Random Forest compare?

• Which evaluation metrics are important when churn cases are relatively uncommon?

4. DATA DISCUSSION

4.1 Data Source and Nature

The project uses a synthetic dataset of 600 customer records generated for academic demonstration. It is not a real telecom company dataset and does not contain personally identifying customer information. The churn label is generated using a simulated probability structure based on selected customer characteristics.

4.2 Dataset Structure

Variable	Description	Type
Customer_ID	Unique synthetic customer identifier	Categorical ID
Tenure_Months	Length of service in months	Numeric
Monthly_Charges	Monthly service charge	Numeric
Contract_Type	Monthly, one-year, or two-year	Categorical
Internet_Service	Fiber, DSL, or none	Categorical
Tech_Support	Whether support service is subscribed	Categorical
Payment_Method	Payment channel	Categorical
Monthly_Usage_Hours	Estimated monthly usage	Numeric
Complaints_Last_6M	Complaints in previous six months	Numeric
Late_Payment	Whether late payment is recorded	Categorical
Churn	Target label: 1 churned, 0 retained	Binary

4.3 Target Variable

The target variable is **Churn**. A value of 1 indicates a simulated churn event and a value of 0 indicates a retained customer. The model learns from the relationship between this target and the input variables.

4.4 Data Preparation

• Customer_ID is excluded from model training because it is an identifier.

• Numeric variables are imputed using the median and standardized.

• Categorical variables are imputed using the most frequent value and one-hot encoded.

• The dataset is divided into training and testing subsets using stratified sampling.

• The test set is held out for evaluation and is not used to fit the models.

5. RESEARCH METHODOLOGY

5.1 Research Design

The project follows a quantitative, model-based predictive approach. First, the synthetic dataset is reviewed and prepared. Next, the data is divided into training and test sets. Two classification algorithms are fitted using the training data. Finally, predictions on the test set are compared using multiple evaluation measures.

5.2 Train-Test Split

A 75:25 split is used. Approximately 75% of records are used to train the models and 25% are reserved for testing. Stratification is applied so that the proportion of churn and non-churn labels is maintained as closely as possible in both subsets.

5.3 Algorithms Selected

A. Logistic Regression

Logistic Regression is a statistical classification algorithm that estimates the probability of a binary outcome. It is relatively interpretable and provides a useful baseline for churn prediction. The predicted probability can be converted into a class using a selected decision threshold.

B. Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees. Each tree is trained using a sample of the data and a subset of features. The combined predictions can capture non-linear patterns and interactions between customer attributes.

5.4 Class Imbalance Consideration

Churn cases are less frequent than retained customers in this synthetic dataset. Class imbalance can make accuracy appear strong even when a model misses many churn cases. For this reason, precision, recall, F1-score, and ROC-AUC are reported in addition to accuracy.

6. TOOLS AND TECHNOLOGIES

Tool / Technology	Purpose
Python	Data handling and model development
Pandas	Dataset creation, cleaning, and grouping
NumPy	Numerical calculations and synthetic data generation
Scikit-learn	Preprocessing, model training, and evaluation
Matplotlib	Charts and visual presentation
Jupyter Notebook	Interactive Python environment
Microsoft Excel	Optional review of tabular data and summary values

6.1 Analytical Techniques

• Frequency counts and percentages for churn distribution.

• Group-wise churn rate comparisons.

• Categorical encoding and numeric scaling.

• Binary classification using two algorithms.

• Confusion matrix and classification metrics.

• ROC-AUC for comparing probability-based ranking performance.

7. WORKFLOW AND PROCEDURE

Step	Activity	Output
1	Define churn outcome and objectives	Project scope
2	Generate structured customer data	600 records
3	Inspect and prepare features	Model-ready inputs
4	Split training and testing data	75:25 split
5	Fit Logistic Regression	Baseline classifier
6	Fit Random Forest	Ensemble classifier
7	Generate test predictions	Predicted labels and probabilities
8	Evaluate model performance	Metric comparison
9	Interpret and recommend	Retention insights

7.1 Model Development Procedure

A preprocessing pipeline is used to keep data transformations consistent. Numeric columns are median-imputed and standardized. Categorical columns are imputed and one-hot encoded. The transformed features are passed to each classifier. Using a pipeline reduces the risk of applying different preprocessing steps to training and testing data.

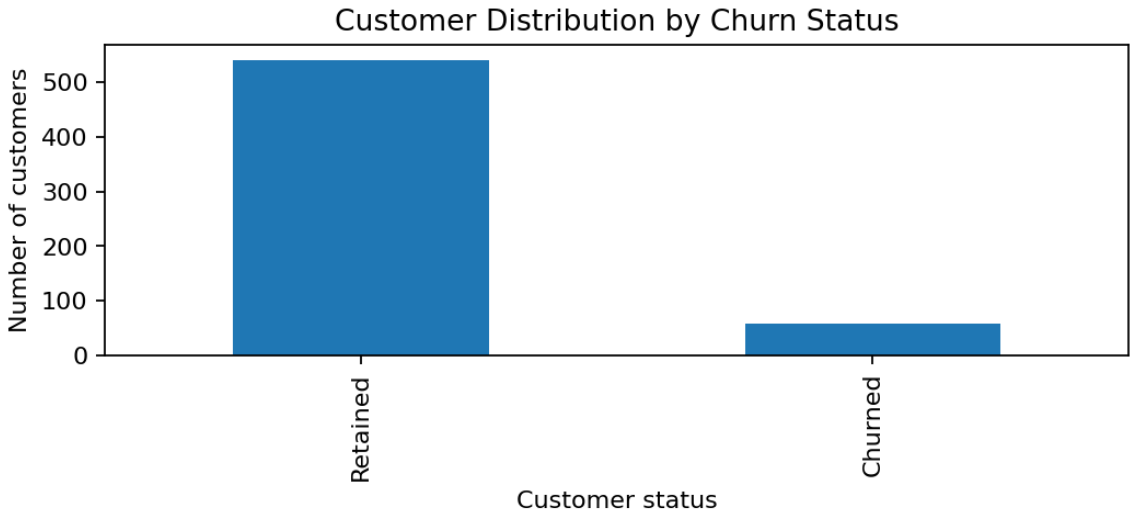
7.2 Evaluation Procedure

Each model predicts churn labels and churn probabilities for the held-out test records. The predictions are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The default classification threshold is retained for the reported precision, recall, and F1-score; a business deployment would require threshold selection based on intervention capacity and the cost of missed churn cases.

8. EXPLORATORY DATA ANALYSIS

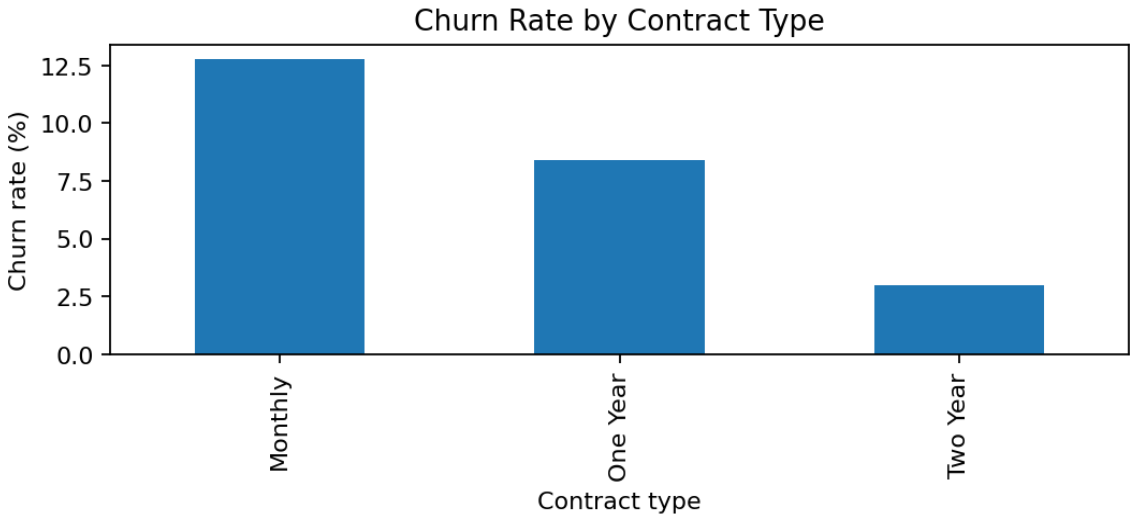
8.1 Overall Churn Distribution

The dataset contains 600 customer records. A total of 59 records (9.83%) are labelled as churned, while 541 records are labelled as retained. The distribution indicates that the target classes are imbalanced in this generated sample.



8.2 Churn by Contract Type

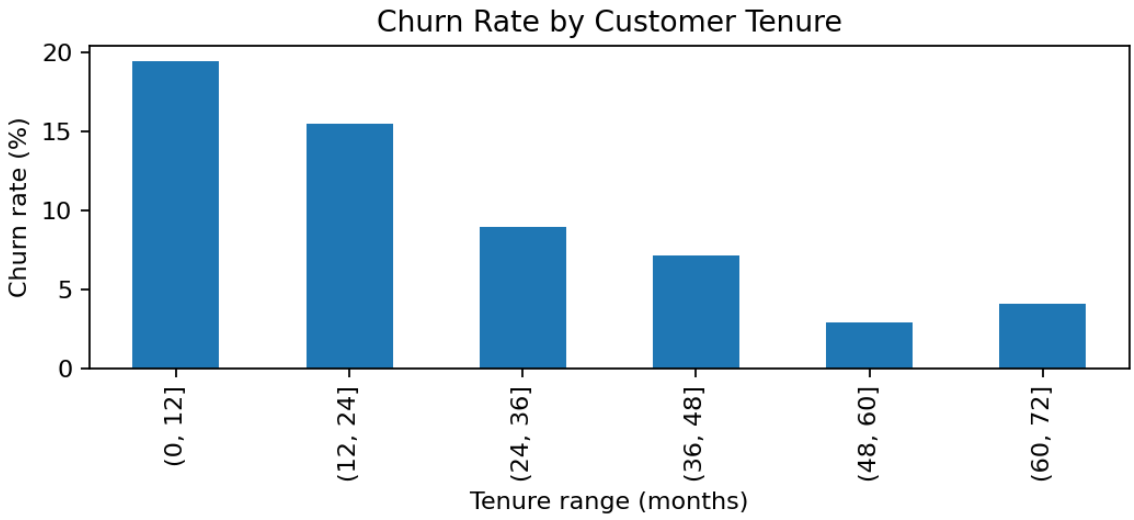
Contract Type	Customers	Churn Rate
Monthly	321	12.77%
One Year	179	8.38%
Two Year	100	3.00%



Contract type is a useful descriptive dimension because customer commitment periods differ. The chart compares the observed sample churn proportion for each contract group; it does not establish that contract length alone causes churn.

8.3 Churn by Tenure

Tenure Group	Customers	Churn Rate
1–12	103	19.42%
13–24	110	15.45%
25–36	101	8.91%
37–48	84	7.14%
49–60	104	2.88%
61–72	98	4.08%



Tenure groups help show how churn labels are distributed across customer relationship stages. Early-tenure customers may require onboarding and service-quality attention, while long-tenure customers may respond to loyalty and value programmes.

8.4 Descriptive Summary of Numeric Variables

Variable	Mean	Std. Dev.	Minimum	Maximum
Tenure Months	35.92	20.97	1.00	72.00
Monthly Charges	1366.13	634.89	309.19	2494.46
Monthly Usage Hours	41.45	14.96	2.00	83.50
Complaints Last 6M	1.07	1.00	0.00	5.00

9. MODEL BUILDING

9.1 Input and Output

Input variables include tenure, monthly charges, contract type, internet service, technical support, payment method, monthly usage, complaint count, and late-payment status. The output is the binary churn label.

9.2 Logistic Regression Model

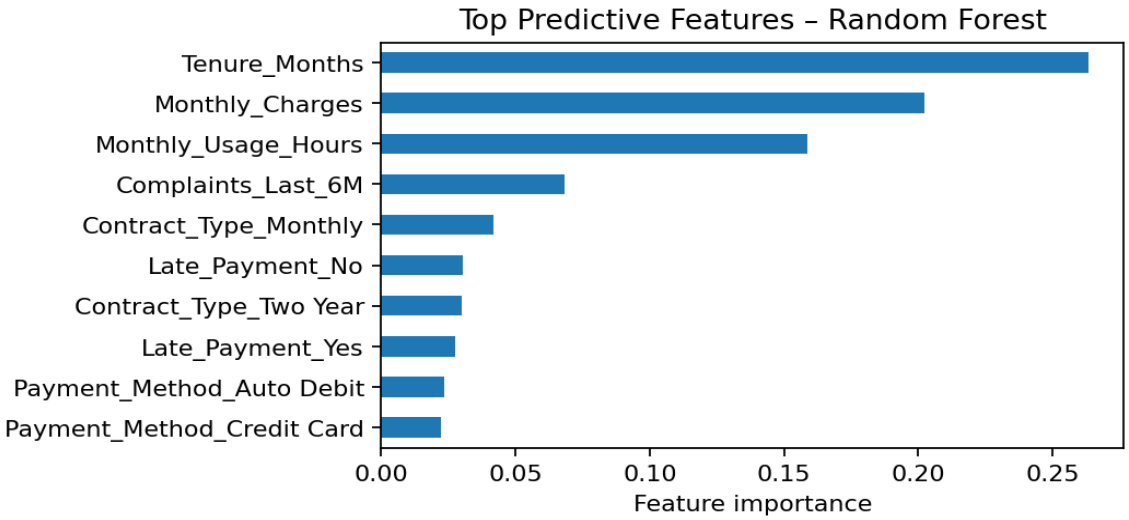
The Logistic Regression classifier is trained with balanced class weights to give additional consideration to the minority class during fitting. It provides a baseline against which the more flexible Random Forest model can be compared.

9.3 Random Forest Model

The Random Forest classifier uses 250 decision trees with a maximum tree depth of 6 and a minimum of 3 samples per leaf. Balanced class weights are used. These settings are demonstration choices rather than an optimized production configuration.

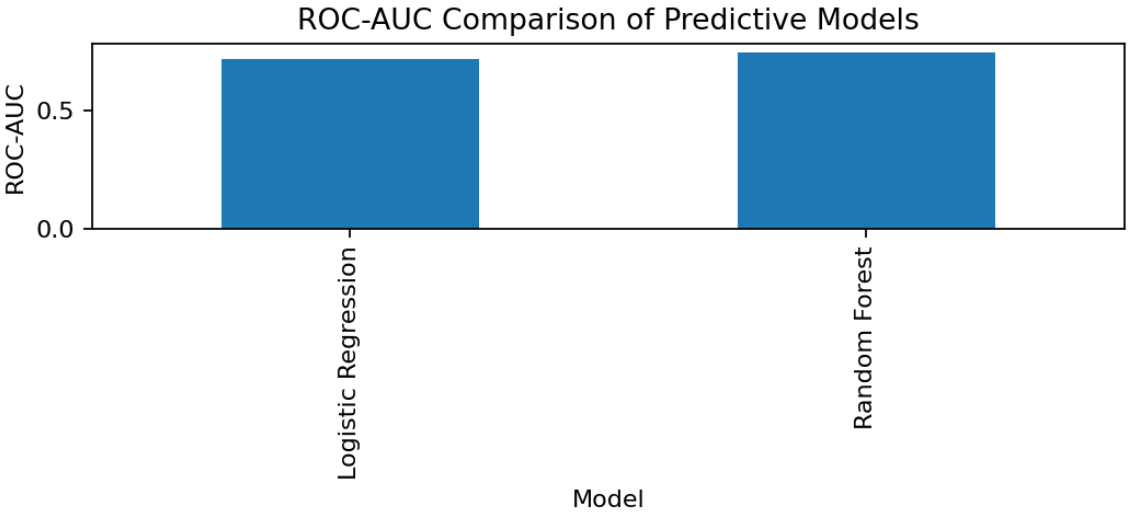
9.4 Feature Importance

Random Forest feature importance summarizes how much each input contributes to reducing impurity across the fitted trees. It can help guide further investigation, but it should not be interpreted as proof of causation or as a direct measure of business impact.



9.5 Model Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.700	0.188	0.600	0.286	0.718
Random Forest	0.880	0.286	0.133	0.182	0.742



10. MODEL EVALUATION AND RESULTS

10.1 Evaluation Metrics

Metric	Meaning	Interpretation
Accuracy	Correct predictions ÷ all predictions	Overall correctness; can be misleading with imbalance
Precision	True churn predictions ÷ all predicted churn	How many flagged customers actually churned
Recall	True churn predictions ÷ all actual churn	How many churn cases the model identified
F1-score	Harmonic mean of precision and recall	Balances precision and recall
ROC-AUC	Area under ROC curve	Measures ranking ability across thresholds

10.2 Best Model in This Experiment

Based on ROC-AUC, **Random Forest** performs best among the two tested models, with a test ROC-AUC of **0.742**. Its test accuracy is 0.880, precision is 0.286, recall is 0.133, and F1-score is 0.182.

10.3 Interpretation

The model has some ability to rank customers by estimated churn risk, but its recall at the default threshold is limited. This means many actual churn cases are not identified by the current classification threshold. Accuracy alone would give an overly positive impression. Before practical use, the model would need further validation, threshold tuning, and a review of the costs associated with missed churn cases and unnecessary retention offers.

10.4 Confusion Matrix

Actual / Predicted	Retained (0)	Churned (1)
Retained (0)	130	5
Churned (1)	13	2

True negatives are retained customers correctly classified as retained. False positives are retained customers incorrectly flagged as churn risks. False negatives are churned customers missed by the model. True positives are churned customers correctly identified.

10.5 Business Meaning

A churn-risk score can help a customer success team prioritize a review of service issues, complaints, payment difficulties, or contract concerns. The score should be combined with customer context and human review. It should not be used to deny service, apply penalties, or make consequential decisions without a fair and transparent process.

11. OBSERVATIONS

• The generated sample includes 600 customer records, with a churn label rate of 9.83%.

• The target variable is imbalanced, so accuracy alone is not sufficient for evaluation.

• Contract type and tenure provide useful dimensions for descriptive churn comparisons.

• The two models produce different precision-recall trade-offs.

• The Random Forest model has the higher ROC-AUC in this experiment.

• The selected model's recall is limited at the default threshold, so many churn cases may remain undetected.

• The dataset is synthetic; the reported metrics are not evidence of real-world predictive performance.

12. RECOMMENDATIONS

• Collect reliable, consented customer data with a clearly defined churn label and observation window.

• Monitor complaints, service quality, payment friction, and early-tenure experience.

• Evaluate recall and precision together rather than selecting a model based only on accuracy.

• Tune the decision threshold according to retention-team capacity and the relative cost of false negatives and false positives.

• Validate models on later time periods to test whether performance remains stable.

• Run controlled retention experiments to measure whether interventions actually reduce churn.

• Explain model outputs to staff and keep human review in the decision process.

• Monitor for data drift, unfair outcomes, and changes in customer behaviour.

13. CONCLUSION

This project demonstrates a complete introductory predictive analytics workflow for customer churn prediction. A synthetic dataset was prepared, explored, split into training and testing subsets, and used to train Logistic Regression and Random Forest classifiers. The models were evaluated with several metrics, including precision, recall, F1-score, and ROC-AUC. Random Forest achieved the highest ROC-AUC in this particular experiment, although its recall at the default threshold remained limited. The project highlights the importance of selecting appropriate evaluation metrics and interpreting model outputs carefully. Predictive analytics can support customer retention planning when it is based on reliable data, validated models, and responsible business processes.

14. LIMITATIONS AND FUTURE SCOPE

14.1 Limitations

• The dataset is synthetic and contains only 600 records.

• The churn labels are generated from an illustrative probability formula.

• The project uses a single random train-test split rather than repeated cross-validation.

• Hyperparameters and decision thresholds are not exhaustively optimized.

• No real campaign outcomes or retention intervention results are included.

• Model performance may change substantially on real customer data.

14.2 Future Scope

• Use a larger, real, authorized customer dataset.

• Apply cross-validation and time-based validation.

• Compare additional algorithms such as Gradient Boosting and XGBoost where available.

• Tune thresholds and calibration to suit business costs.

• Add customer service history, product usage trends, and cancellation reasons where appropriate.

• Measure the business impact of retention interventions through controlled experiments.

15. REFERENCES

• Scikit-learn documentation – Supervised learning and model evaluation: https://scikit-learn.org/stable/supervised_learning.html

• Scikit-learn documentation – Classification metrics: https://scikit-learn.org/stable/modules/model_evaluation.html

• Pandas documentation: https://pandas.pydata.org/docs/

• NumPy documentation: https://numpy.org/doc/

• Matplotlib documentation: https://matplotlib.org/stable/

Note: The references are general learning resources. All customer records and numerical findings in this report were generated for academic demonstration.

16. APPENDIX: SAMPLE DATASET AND PYTHON LOGIC

The following table displays a preview of the synthetic dataset. The complete analysis uses all 600 records.

Customer ID	Tenure	Charges	Contract	Support	Complaints	Late Pay	Churn
C0001	52	2377	Monthly	Yes	0	No	0
C0002	15	2168	Two Year	No	1	Yes	0
C0003	72	843	Monthly	Yes	2	No	0
C0004	61	1290	Monthly	Yes	2	No	0
C0005	21	583	One Year	Yes	2	No	0

Customer ID	Tenure	Charges	Contract	Support	Complaints	Late Pay	Churn
C0006	24	2398	One Year	No	0	Yes	0
C0007	3	1633	Monthly	Yes	0	No	0
C0008	22	802	One Year	Yes	0	No	0
C0009	53	1777	One Year	Yes	1	Yes	0
C0010	2	1659	Monthly	No	0	Yes	0
C0011	30	1087	Two Year	No	4	Yes	0
C0012	38	549	One Year	No	1	No	0
C0013	2	1776	One Year	Yes	2	No	0
C0014	64	1444	Two Year	No	1	No	0
C0015	60	1998	Two Year	Yes	2	Yes	0

16.1 Illustrative Python Workflow

```
# Split the data
X = data.drop(columns=["Customer_ID", "Churn"])
y = data["Churn"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

# Fit a classification pipeline
model = Pipeline([
    ("preprocessing", preprocessor),
    ("classifier", RandomForestClassifier(
        n_estimators=250, max_depth=6,
        class_weight="balanced", random_state=42
    ))
])
model.fit(X_train, y_train)

# Evaluate on held-out test data
predictions = model.predict(X_test)
probabilities = model.predict_proba(X_test)[:, 1]
```

Academic integrity note: This is an illustrative project built using synthetic data. Replace the sample data and rerun the workflow if actual organizational analysis is required.